INCH-POUND

MIL-DTL-3965/19F
19 August 2002
SUPERSEDING
MIL-C-3965/19E
26 August 1970

DETAIL SPECIFICATION SHEET
CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM,
(POLARIZED, SINTERED SLUG) - 85°C (VOLTAGE DERATED TO 125°C),
STYLES CL17 AND CL18

INACTIVE FOR NEW EQUIPMENT DESIGN
NO SUPERSEDING DOCUMENT

This specification is approved for use to the Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-DTL-3965.

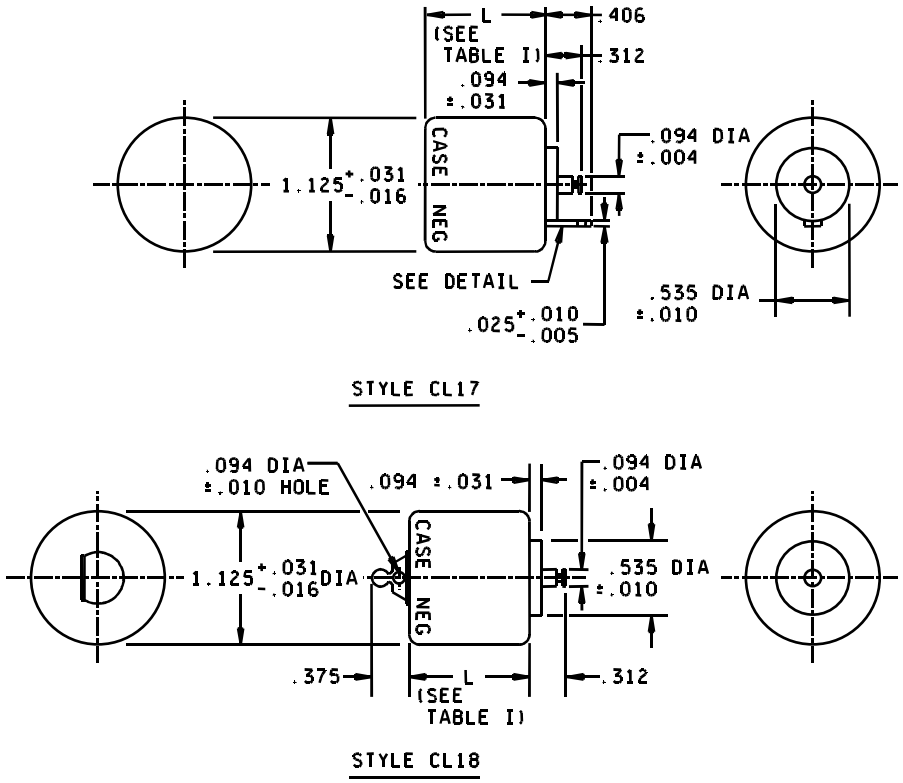
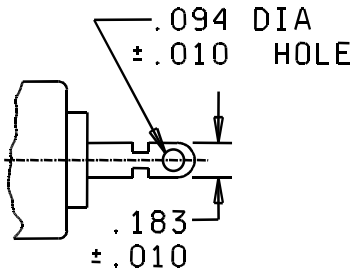


FIGURE 1. Styles CL10 and CL13 capacitors.

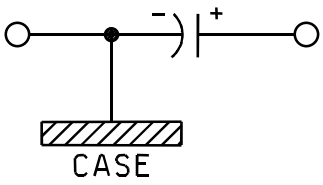
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DETAIL OF CL 17

SOLDER-LUG TERMINAL



CIRCUIT DIAGRAM

Inches	mm	Inches	mm
.004	0.10	.183	4.65
.005	0.13	.312	7.92
.010	0.25	.375	9.53
.016	0.41	.406	10.31
.025	0.64	.535	13.59
.031	0.79	1.125	28.58
.094	2.39		

NOTES:

1. All dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ± .062 (1.57 mm).

FIGURE 1. Styles CL17 and CL18 capacitors – Continued.

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TABLE I. Capacitor characteristics.

PIN 1/	Voltage			Capaci-tance	Cap. toler-ance	Dissi-pation factor (max)	DC leakage (max)			Imped-ance (max)	Cap. change (-55°C)	Dimension L ±.062 (1.57)
	Rated (+85°C)	Derated (+125°C)	Surge (+85°C)				+25°C	+85°C	+125°C			
	volts, dc	volts, dc	volts,dc	µF		%	µA	µA	µA	ohms	%	Inches
CL1-BH371-PG	30	26	34.5	370	M, T	41.0	18	125	180	15	-65	.600(15.24)
CL1-BH651-PG	30	26	34.5	650	M, T	74.0	21	145	210	15	-85	1.100(27.94)
CL1-BH132-PG	30	26	34.5	1,300	M, T	148.0	27	190	270	10	-85	1.100(27.94)
CL1-BK201-PG	60	52	69.0	200	M, T	22.5	19	135	190	30	-50	.600(15.24)
CL1-BK351-PG	60	52	69.0	350	M, T	39.8	22	155	220	25	-70	1.100(27.94)
CL1-BK701-PG	60	52	69.0	700	M, T	79.0	29	200	290	15	-70	1.100(27.94)
CL1-BM121-PG	90	78	103.0	120	M, T	13.0	19	135	190	40	-40	.600(15.24)
CL1-BM221-PG	90	78	103.0	220	M, T	24.8	21	145	210	30	-60	1.100(27.94)
CL1-BM451-PG	90	78	103.0	450	M, T	50.2	29	195	290	35	-60	1.100(27.94)
CL1-BR420-PG	180	155	207.0	42	M, T	16.1	17	120	170	75	-40	.976(24.79)
CL1-BR600-PG	180	155	207.0	60	M, T	13.0	19	135	190	60	-40	.976(24.79)
CL1-BR111-PG	180	155	207.0	110	M, T	24.8	21	145	210	60	-60	1.938(49.23)
CL1-BR231-PG	180	155	207.0	230	M, T	52.0	29	200	290	50	-60	1.938(49.23)
CL1-BS280-PG	270	235	310.0	28	M, T	16.1	19	120	190	80	-40	1.350(34.29)
CL1-BS400-PG	270	235	310.0	40	M, T	22.5	19	135	190	100	-40	1.350(34.29)
CL1-BS750-PG	270	235	310.0	75	M, T	24.8	21	145	210	90	-60	2.812(71.42)
CL1-BS151-PG	270	235	310.0	150	M, T	52.0	28	195	280	75	-60	2.812(71.42)
CL1-BT220-PG	360	310	414.0	22	M, T	16.1	18	125	180	100	-40	1.705(43.31)
CL1-BT300-PG	360	310	414.0	30	M, T	22.5	19	135	190	120	-40	1.705(43.31)
CL1-BU170-PG	450	390	518.0	17	M, T	16.1	18	125	180	130	-40	2.080(52.83)
CL1-BU250-PG	450	390	518.0	25	M, T	23.5	19	135	190	150	-40	2.080(52.83)
CL1-BV140-PG	540	470	621.0	14	M, T	16.1	17	120	170	160	-40	2.435(61.85)
CL1-BV200-PG	540	470	621.0	20	M, T	22.5	19	135	190	170	-40	2.435(61.85)
CL1-BW120-PG	630	545	724.0	12	M, T	16.1	17	120	170	180	-40	2.810(71.37)
CL1-BW180-PG	630	545	724.0	18	M, T	23.5	19	135	190	200	-40	2.810(71.37)

1/ Complete type designation will include additional symbols to indicate style and capacitance tolerance.

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TABLE II. Cross-reference data.

From	To	From	To
PIN 1/ MIL-DTL-3965/19	PIN 2/ MIL-PRF-39006/20	PIN 1/ MIL-DTL-3965/19	PIN 2/ MIL-PRF-39006/20
	M39006/20-		M39006/20-
CL1-BH371MPG	1000	CL1-BR231TPG	1025
CL1-BH371TPG	1001	CL1-BS280MPG	1026
CL1-BH651MPG	1002	CL1-BS280TPG	1027
CL1-BH651TPG	1003	CL1-BS400MPG	1028
CL1-BH132MPG	1004	CL1-BS400TPG	1029
CL1-BH132TPG	1005	CL1-BS750MPG	1030
CL1-BK201MPG	1006	CL1-BS750TPG	1031
CL1-BK201TPG	1007	CL1-BS151MPG	1032
CL1-BK351MPG	1008	CL1-BS151TPG	1033
CL1-BK351TPG	1009	CL1-BT220MPG	1034
CL1-BK701MPG	1010	CL1-BT220TPG	1035
CL1-BK701TPG	1011	CL1-BT300MPG	1036
CL1-BM121MPG	1012	CL1-BT300TPG	1037
CL1-BM121TPG	1013	CL1-BU170MPG	1038
CL1-BM221MPG	1014	CL1-BU170TPG	1039
CL1-BM221TPG	1015	CL1-BU250MPG	1040
CL1-BM451MPG	1016	CL1-BU250TPG	1041
CL1-BM451TPG	1017	CL1-BV140MPG	1042
CL1-BR420MPG	1018	CL1-BV140TPG	1043
CL1-BR420TPG	1019	CL1-BV200MPG	1044
CL1-BR600MPG	1020	CL1-BV200TPG	1045
CL1-BR600TPG	1021	CL1-BW120MPG	1046
CL1-BR111MPG	1022	CL1-BW120TPG	1047
CL1-BR111TPG	1023	CL1-BW180MPG	1048
CL1-BR231MPG	1024	CL1-BW180TPG	1049

1/ Complete PIN will include an additional symbol to indicate style.
2/ PIN referenced will substitute for styles CL17 and CL18.

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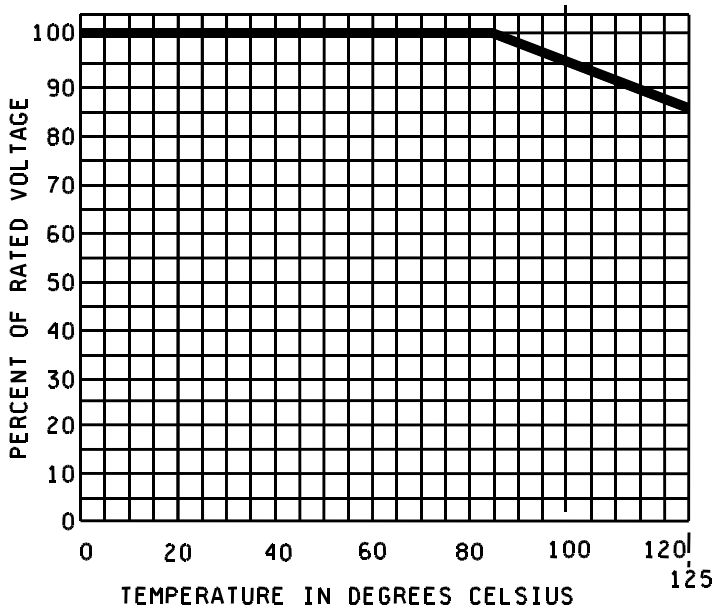


FIGURE 2. Voltage derating with temperature.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type: Tubular; uninsulated.

Seal type: Hermetic (symbol G).

Terminals: Solder-lug.

Rated temperature: -55°C to +85°C, voltage derated to +125°C (see figure 2).

Rated voltage: See table I.

Capacitance (cap) (nom): See table I.

Capacitance tolerance: Symbol M (± 20 percent) or T (+50 percent, -15 percent).

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DC leakage (DCL) (max) (at +25°C): See table I.

Dissipation factor (DF) (max): See table I.

Seal: In accordance with MIL-DTL-3965.

Dielectric withstanding voltage (DWV):

Sleeving to case (min): Not applicable.

Terminal to case: Not applicable.

Insulation resistance:

Sleeving to case (min): Not applicable.

Terminal to case: Not applicable.

Temperature cycling:

DCL (max): 125 percent of 25°C value of table I.

ΔCap.: Within $\pm$ 5 percent of initial measurement.

DF (max): 115 percent of table I value.

Solderability: Not applicable.

Terminal strength:

Pull test: In accordance with MIL-DTL-3965.

Bend test (if required): In accordance with MIL-DTL-3965.

Torque test: Not applicable.

Wire-lead bend test: Not applicable.

Twist test: Not applicable.

Surge voltage: See table I.

Moisture resistance:

DCL (max): 125 percent of 25°C value of table I.

ΔCap.: Within $\pm$ 8 percent of initial measurement.

DF (max): 115 percent of table I value.

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Low temperature (storage):

DCL (max): See table I (25°C).

ΔCap.: Within ± 5 percent of initial measurement.

DF (max): See table I.

Stability at low and high temperature:

Step 1 (+25°C):

DCL (max): See table I.

Cap.: Within tolerance of table I value.

DF (max): See table I.

Step 2 (-55°C):

Impedance: See table I.

ΔCap. (max): Within percent shown in table I of step 1 value.

Step 3 (+25°C):

DCL (max): See table I.

ΔCap.: Within ±5 percent of step 1 value.

DF (max): See table I.

Step 4 (+85°C):

DCL (max): See table I.

ΔCap (max): +20 percent of step 1 value.

DF (max): See table I.

Step 5 (+125°C):

DCL (max): See table I.

ΔCap. (max): +25 percent of step 1 value.

DF (max): See table I.

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Step 6 (+25°C):

DCL (max): See table I.

ΔCap.: Within ±5 percent of initial measurement.

DF (max): See table I.

Reverse voltage: Not applicable.

Life at +85°C:

Qualification (2,000 hours):

DCL (max): See table I (25°C).

ΔCap.: Within ±20 percent of initial measurement.

DF (max): See table I.

Quality conformance:

240 hours:

DCL (max): See table I (25°C).

ΔCap.: Within ±10 percent of initial measurement.

DF (max): See table I.

1,760 hours:

DCL (max): See table I (25°C).

ΔCap.: Within ±20 percent of initial measurement.

DF (max): See table I.

Barometric pressure:

Test voltage: Rated voltage for 30 V to 360 V units; 500 V dc for 540 V and 630 V.

Life at +125°C.

DCL (max): See table I (25°C).

ΔCap.: Within ±20 percent of initial measurement.

DF (max): 150 percent of initial requirement.

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DWV and IR: Not applicable.

Marking: Capacitors shall be marked as shown in the following example:

EXAMPLE:

CL17BH371MPG	-	Type designation
12345 7015	-	Source and date code
370 μF 30 V dc	-	Capacitance and rated voltage (at 85°C).

Polarity: Capacitors shall be marked “CASE NEG” on the negative end (see figure 1).

Voltage groups (85°C rated voltage):

I - - - - - - - - - -	30 V to 90 V inclusive
II - - - - - - - - - - -	180 to 360 V, inclusive
III- - - - - - - - - -	450 V to 630 V, inclusive.

NOTE: The capacitors covered by this specification sheet are substitutable for those covered by MIL-C-3965/19. See table II.

Custodians:
Army – CR
Navy – EC
Air Force – 11
DLA - CC

Preparing activity:
DLA - CC

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